Spontaneous Symmetry Breaking

Introducing the Problem

Let's start by looking at the φ^4 theory:

$\mathcal{L} = -\frac{1}{2}\partial^\mu\varphi\partial_\mu\varphi - \frac{1}{2}m^2\varphi^2 - \frac{1}{24}\lambda\varphi^4$ $\longrightarrow$ Obey's a Z_2 symmetry. Lagrangian is invariant under:

$$\varphi(x) \longrightarrow -\varphi(x)$$

The unitary operator Z does this:

$Z^{-1}\varphi(x)Z = -\varphi(x)$

Due to unitarity.

We will have: $Z^2 = 1 \longrightarrow Z^{-1} = Z$, $Z^{-1} = Z^\dagger$. This makes the operator hermitian.

Now, assume we have negative mass: $m^2 < 0$. We can write $\mathcal{L}$ as:

$\mathcal{L} = -\frac{1}{2}\partial^\mu\varphi\partial_\mu\varphi - V(\varphi)$; $V(\varphi) = \frac{1}{2}m^2\varphi^2 + \frac{1}{24}\lambda\varphi^4 = \frac{1}{24}\lambda(\varphi^2 - v^2) - \frac{1}{24}\lambda v^4$

Here, we have defined: $v \equiv (+6|m^2|/\lambda)^{1/2}$

Since the last term will essentially be inverse order of the coupling, we can drop that.

Now, there are two classical field configurations that minimize the energy of the system: $\varphi(x) = \pm v$

This should carry forward in the Quantum theory. So we can define two ground states as:

$\langle 0+|\varphi(x)|0+\rangle = +v$; $\langle 0-|\varphi(x)|0-\rangle = -v$ $\longrightarrow$ At tree-level (no loop corrections yet)

The two ground states will be exchanged by the operator Z:

$Z|0+\rangle = |0-\rangle$

* verification: $\langle 0+|\varphi(x)|0+\rangle = \langle 0+|Z^\dagger - \varphi(x)Z|0+\rangle = (\langle 0+|Z^\dagger)[-\varphi(x)](Z|0+\rangle) = v$

$\longrightarrow (\langle 0+|Z^\dagger)(\varphi(x))(Z|0+\rangle) = -v = \langle 0-|\varphi(x)|0-\rangle$ Hence proved.

These states will also be orthogonal: $\langle 0+|0-\rangle = 0$

Why? Consider a single particle QM problem: $H = \frac{1}{2}p^2 + \frac{1}{24}\lambda(x^2 - v^2)^2$

This has two ground states with: $\varphi_\pm(x) = \langle x|0\pm\rangle \sim \exp[-\omega(x \mp v)^2/2]$; $\omega = (\lambda v^2/3)^{1/2}$

The true ground state is a symmetric linear combination of these. Anti-symmetric one has a slightly higher energy due to tunneling.

Now, we can think of a field theory as an infinite set of oscillators which are coupled together. Now, for the field to go from a $|0+\rangle$ to a $|0-\rangle$ state, all infinite of these oscillators have to tunnel. Since there are infinite such oscillators, their amplitudes all get raised to ∞ power and vanish.

This implies we can only pick one ground state: either $|0+\rangle$ or $|0-\rangle$. We will work with $|0+\rangle$.

We can then define a shifted field: $\rho(x) = \varphi(x) - v \longrightarrow \langle 0+|\rho(x)|0+\rangle = 0$

The potential then becomes: $V(\varphi) = \frac{1}{24}\lambda[(\rho+v)^2 - v^2]^2 = \frac{1}{6}\lambda v^2\rho^2 + \frac{1}{6}\lambda v\rho^3 + \frac{1}{24}\lambda\rho^4$

We have the lagrangian:

$\mathcal{L} = -\frac{1}{2}\partial^\mu\rho\,\partial_\mu\rho - \frac{1}{6}\lambda v^2\rho^2 - \frac{1}{6}\lambda\rho^3 - \frac{1}{24}\lambda\rho^4 \longrightarrow$ we can see that: $\frac{1}{6}\lambda v^2 = |m^2| = \frac{1}{2}m_\rho^2$

This here is a completely valid QFT but, it does not hold the same Z_2 symmetry. Then we say that the Z_2 symmetry is spontaneously broken.

Renormalization and the Quantum Action

So, how does renormalization work here? Look at the original lagrangian:

$$\mathcal{L} = -\frac{1}{2}Z_\varphi\partial^\mu\varphi\,\partial_\mu\varphi - \frac{1}{2}Z_m m^2\varphi^2 - \frac{1}{24}Z_\lambda\lambda\varphi^4$$

For the positive mass, this renormalizes just fine. However, for the negative mass and the shifted lagrangian, we seem to need another counter-term for the cubic interaction and one to remove the tadpoles.

So, how can we make do with just the results from our original Feynman diagrams?

To see this, consider the original Quantum action:

$$\Gamma(\varphi) = \frac{1}{2}\int\frac{d^4k}{(2\pi)^4}\,\tilde{\varphi}(-k)(k^2+m^2-\Pi(k^2))\tilde{\varphi}(k) + \sum_{n=3}^{\infty}\frac{1}{n!}\int\frac{d^4k_1}{(2\pi)^4}\cdots\frac{d^4k_n}{(2\pi)^4}(2\pi)^4\delta^4(k_1+\dots+k_n)$$
$$\times V_n(k_1,\dots,k_n)\tilde{\varphi}(k_1)\dots\tilde{\varphi}(k_n)$$

computed for positive mass.

The self energy, vertices etc. are all made finite by a suitable renormalization scheme.

This QA is made finite via 3 counter-terms and retains the symmetry of the classical action.

Symmetry has to be linear (transformed fields are linear f^n's of original) and the same symmetry needs to be obeyed by $\mathcal{D}\varphi$. Any symmetry where $\mathcal{D}\varphi$ is not invariant is called anomalous.

With the positive mass case accounted for, let's look at the negative one

Quantum E.O.M in presence of a source: $\frac{\delta\Gamma}{\delta\varphi(x)} = -J(x)$; Solution: Vacuum expectation value.

Set source to 0: $\frac{\delta\Gamma}{\delta\varphi(x)} = 0$; and look for a translationally invariant solution $\varphi(x) = v$.

This is equivalent to minimizing the quantum potential:

$$\Gamma(\varphi) = \int d^4x\left[-U(\varphi) - \tfrac{1}{2}Z(\varphi)\partial^\mu\varphi\,\partial_\mu\varphi + \dots\right]$$

For a weakly coupled theory, the quantum potential will be qualitatively similar to the classical potential.

Thus, negative mass will imply all our previous observations and show that even the Quantum action exhibits symmetry breaking.

Since Q.A is always finite, we don't have to worry about renormalization.